

ZenVoice Integration Guide

AWS Nova Sonic 2 & ElevenLabs TTS

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1. Overview

This document describes how two new voice providers have been integrated into the ZenVoice voice agent platform:

AWS Nova Sonic 2 — A realtime speech-to-speech model from Amazon Bedrock. It replaces the entire STT + LLM + TTS pipeline with a single bidirectional audio model (same pattern as Gemini Realtime).

ElevenLabs TTS — A high-quality text-to-speech provider used in the pipelined approach (STT + LLM + ElevenLabs TTS).

2. Architecture

The agent supports two operational modes depending on the provider selected:

Mode	Provider Name	Pipeline	Components Used
Pipelined	ElevenLabs	STT -> LLM -> TTS	Deepgram STT + OpenAI LLM + ElevenLabs TTS
Realtime	Nova Sonic	Speech-to-Speech	AWS Nova Sonic 2 (single model)
Realtime	Gemini Realtime	Speech-to-Speech	Google Gemini (single model)

3. Prerequisites

3.1 Dependencies

The following packages were added to `pyproject.toml`:

```
livekit-agents[aws,azure,deepgram,elevenlabs,google,groq,lmnt,openai,sarvam,silero]~=1.3.10
livekit-plugins-aws[realtime]
```

Install with: `uv sync`

3.2 Environment Variables

Variable	Provider	Description
ELEVEN_API_KEY	ElevenLabs	API key from elevenlabs.io/app/settings/api-keys
AWS_ACCESS_KEY_ID	Nova Sonic	IAM access key with Bedrock permissions
AWS_SECRET_ACCESS_KEY	Nova Sonic	IAM secret access key

AWS_DEFAULT_REGION	Nova Sonic	AWS region (default: ap-northeast-1)
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IMPORTANT: Nova Sonic 2 is NOT available in ap-south-1. Supported regions: us-east-1, us-west-2, eu-north-1, ap-northeast-1 (Tokyo).

4. AWS Nova Sonic 2 Integration

4.1 What is Nova Sonic 2?

Amazon Nova Sonic 2 is a speech-to-speech model from AWS Bedrock. It handles speech understanding and generation in a single model with bidirectional audio streaming, 16 voices across 8 languages, and a 1M token context window.

4.2 Configuration

The model is created via the LiveKit AWS plugin:

```
from livekit.plugins import aws

model = aws.realtime.RealtimeModel.with_nova_sonic_2(
    voice="tiffany",
    turn_detection="MEDIUM",
    region="ap-northeast-1",
)
```

4.3 Parameters

Parameter	Type	Default	Description
voice	string	tiffany	Voice name (16 available: tiffany, joanna, matthew, aria, etc.)
turn_detection	string	MEDIUM	Turn-taking sensitivity: HIGH, MEDIUM, or LOW
region	string	us-east-1	AWS region for Bedrock endpoint
temperature	float	0.7	Sampling temperature (0-1)
top_p	float	0.9	Nucleus sampling probability
max_tokens	int	1024	Upper bound for tokens emitted
generate_reply_timeout	float	10.0	Timeout for generate_reply() in seconds

Note: Nova Sonic has an 8-minute session limit. The LiveKit plugin handles session recycling automatically.

4.4 API Request Example

```
POST /make_call
{
  "phone_number": "+919876543210",
  "outbound_trunk_id": "ST_xxx",
  "realtime_provider_name": "Nova Sonic",
  "realtime_model_config": {
    "voice": "tiffany",
    "turn_detection": "MEDIUM"
  },
  "instructions": "You are a helpful assistant.",
  "first_message": "Hello! How can I help?"
}
```

5. ElevenLabs TTS Integration

5.1 What is ElevenLabs?

ElevenLabs provides AI-powered text-to-speech with natural-sounding voices. It is used as a TTS provider in the pipelined approach (STT + LLM + TTS). The LiveKit plugin authenticates via the ELEVEN_API_KEY environment variable.

5.2 Configuration

```
from livekit.plugins import elevenlabs

session = AgentSession(
    tts=elevenlabs.TTS(
        voice_id="ODq5zmih8GrVes37Dizd",
        model="eleven_multilingual_v2"
    ),
    stt=deepgram.STT(),
    llm=openai.LLM(),
    vad=silero.VAD.load(),
)
```

5.3 Parameters

Parameter	Type	Default	Description
voice_id	string	EXAVITQu4vr4xnSDxMaL	Voice ID from ElevenLabs voice library
model	string	eleven_flash_v2_5	TTS model (eleven_flash_v2_5, eleven_multilingual_v2, etc.)
language	string	en	ISO-639-1 language code
stability	float	0.5	Voice stability (0-1, via voice_settings)
similarity_boost	float	0.75	Voice similarity (0-1, via voice_settings)
speed	float	1.0	Speech speed multiplier (via voice_settings)
enable_ssml_parsing	bool	false	Enable SSML markup for pronunciation control

5.4 API Request Example

```
POST /make_call
{
  "phone_number": "+919876543210",
  "outbound_trunk_id": "ST_xxx",
  "tts_provider_name": "ElevenLabs",
  "tts_config": {
    "voice_id": "ODq5zmih8GrVes37Dizd",
    "model": "eleven_multilingual_v2"
  },
  "instructions": "You are a helpful assistant.",
  "first_message": "Hello! How can I help?"
}
```

6. Local Testing

A `generate_token.py` script is provided for quick local testing. It generates a LiveKit token with an embedded agent dispatch:

```
# Start the agent worker first
uv run agents/agent.py dev

# In another terminal, generate token + meet link
uv run python generate_token.py --provider nova-sonic
uv run python generate_token.py --provider elevenlabs
uv run python generate_token.py --provider gemini
uv run python generate_token.py --provider default
```

The script outputs a `meet.livekit.io` URL. Open it in a browser to join the room and talk to the agent.

7. Files Changed

File	Change
pyproject.toml	Added aws, elevenlabs extras + livekit-plugins-aws[realtime]
agents/.env	Added ELEVEN_API_KEY, AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY, AWS
agents/agent.py	Added aws/elevenlabs imports, factory functions, router entries
generate_token.py	New file: token generation with provider presets and RoomAgentDispatch
docs/integration_guide.pdf	This document

8. Supported Providers Summary

Component	Provider	Provider Name Value	Default
Realtime	Gemini	realtime_provider_name="Gemini Realtime"	
Realtime	Nova Sonic 2	realtime_provider_name="Nova Sonic"	
LLM	OpenAI	llm_provider_name="OpenAI"	Yes
LLM	Groq	llm_provider_name="Groq"	
STT	Deepgram	stt_provider_name="Deepgram"	Yes
STT	Sarvam	stt_provider_name="Sarvam"	
STT	Groq	stt_provider_name="Groq"	
STT	Azure	stt_provider_name="Azure"	
TTS	Deepgram	tts_provider_name="Deepgram"	Yes
TTS	ElevenLabs	tts_provider_name="ElevenLabs"	
TTS	Sarvam	tts_provider_name="Sarvam"	
TTS	Gemini	tts_provider_name="Gemini"	
TTS	Groq	tts_provider_name="Groq"	
TTS	Azure	tts_provider_name="Azure"	
TTS	LMNT	tts_provider_name="lmnt"	